

Figure 2 shows the energy levels of an atom.

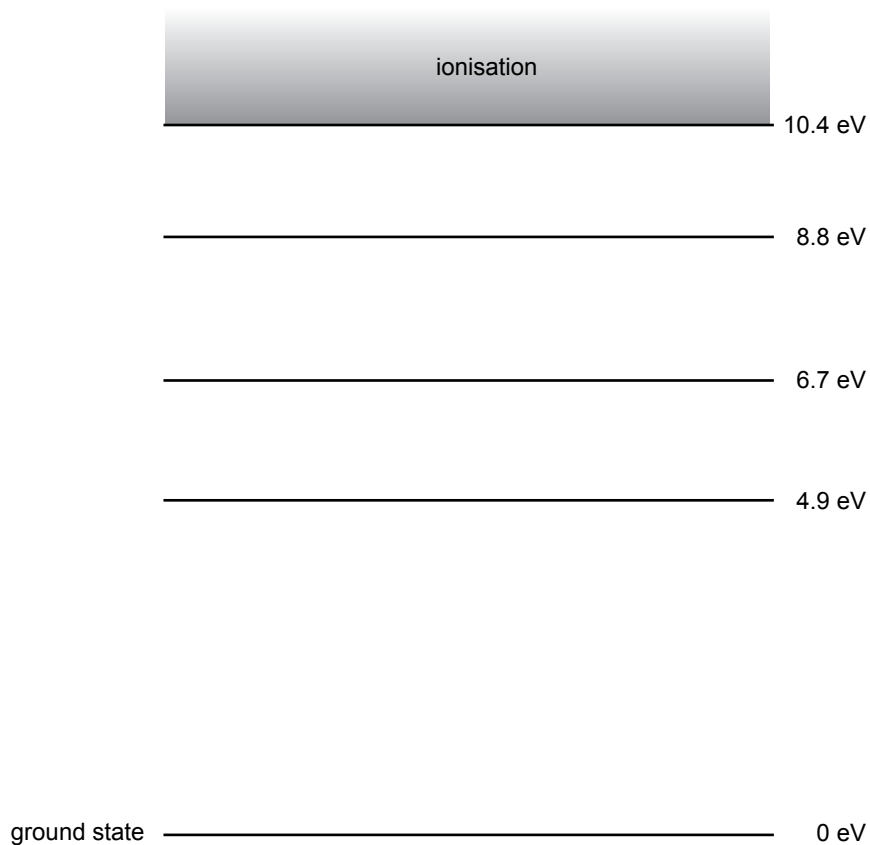


Figure 2

### Question 6

For atoms in the 2nd excited state (6.7 eV), calculate **all** the possible energies of the photons emitted from transitions back to the ground state.

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eV

3 marks

# **PHYSICS**

## **Written examination 2**

### **DATA SHEET**

#### **Directions to students**

Detach this data sheet before commencing the examination.

This data sheet is provided for your reference.

|    |                                                                 |                                                                                     |
|----|-----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1  | velocity; acceleration                                          | $v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$                |
| 2  | equations for constant acceleration                             | $v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$ |
| 3  | Newton's second law                                             | $F = ma$                                                                            |
| 4  | circular motion                                                 | $a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$                                          |
| 5  | Hooke's law                                                     | $F = -kx$                                                                           |
| 6  | elastic potential energy                                        | $\frac{1}{2}kx^2$                                                                   |
| 7  | gravitational potential energy near the surface of the Earth    | $mgh$                                                                               |
| 8  | kinetic energy                                                  | $\frac{1}{2}mv^2$                                                                   |
| 9  | torque                                                          | $\tau = Fr$                                                                         |
| 10 | Newton's law of universal gravitation                           | $F = G \frac{M_1 M_2}{r^2}$                                                         |
| 11 | gravitational field                                             | $g = G \frac{M}{r^2}$                                                               |
| 12 | stress                                                          | $\sigma = \frac{F}{A}$                                                              |
| 13 | strain                                                          | $\varepsilon = \frac{\Delta L}{L}$                                                  |
| 14 | Young's modulus                                                 | $E = \frac{\text{stress}}{\text{strain}}$                                           |
| 15 | electric force on charged particle in an electric field         | $F = qE$                                                                            |
| 16 | electric field between charged plates                           | $E = \frac{V}{d}$                                                                   |
| 17 | energy change of charged particle moving between charged plates | $\Delta E_k = qV$                                                                   |
| 18 | photoelectric effect                                            | $E_{k \max} = hf - W$                                                               |
| 19 | photon energy                                                   | $hf$                                                                                |
| 20 | photon momentum                                                 | $p = \frac{h}{\lambda}$                                                             |
| 21 | de Broglie wavelength                                           | $\lambda = \frac{h}{p}$                                                             |

|                                                      |                                                          |
|------------------------------------------------------|----------------------------------------------------------|
| Gravitational field strength at the surface of Earth | $g = 9.8 \text{ N kg}^{-1}$                              |
| Universal gravitational constant                     | $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ |
| Mass of Earth                                        | $M_E = 5.98 \times 10^{24} \text{ kg}$                   |
| Radius of Earth                                      | $R_E = 6.37 \times 10^6 \text{ m}$                       |
| Mass of the Sun                                      | $M_{\text{SUN}} = 2.0 \times 10^{30} \text{ kg}$         |

|                        |                                                                                   |
|------------------------|-----------------------------------------------------------------------------------|
| Mass of the electron   | $m_e = 9.1 \times 10^{-31} \text{ kg}$                                            |
| Charge on the electron | $e = 1.6 \times 10^{-19} \text{ C}$                                               |
| Planck's constant      | $h = 6.63 \times 10^{-34} \text{ J s}$<br>$h = 4.14 \times 10^{-15} \text{ eV s}$ |
| Speed of light         | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                                            |

#### Prefixes/Units

$$\text{m} = \text{milli} = 10^{-3}$$

$$\mu = \text{micro} = 10^{-6}$$

$$\text{n} = \text{nano} = 10^{-9}$$

$$\text{k} = \text{kilo} = 10^3$$

$$\text{M} = \text{mega} = 10^6$$

$$\text{G} = \text{giga} = 10^9$$

$$\text{tonne} = 10^3 \text{ kg}$$

**END OF DATA SHEET**